MIE375 — Financial Engineering

Course Notes

Bonnie Zhuang

Fall 2026

Contents

Lecture 1: Introduction/Syllabus **2**

Lecture 2: Present Value and Internal Rate of Return **7**

Lecture 3: Fixed Income Securities **14**

Lecture 4: The Term Structure of Interest Rates **21**

Lecture 5: Cash Matching **28**

Lecture 6: Bond Price Sensitivity **32**

Lecture 7: Duration for Bond Portfolios & Immunization **40**

Lecture 1: Introduction/Syllabus

September 14, 2026

Principal and Interest

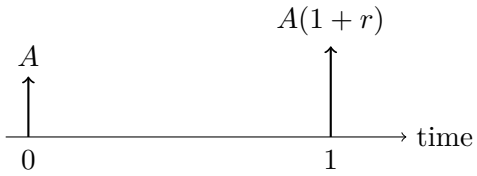
Def. Principal: Amount invested

Interest: "Rent" paid on investment

In general, for

- A: Principal
- r: interest rate per period
- V: Value received.

After 1 period, $V = A(1+r) = A + Ar$ is received, where Ar is the interest earned on the principal for one period.

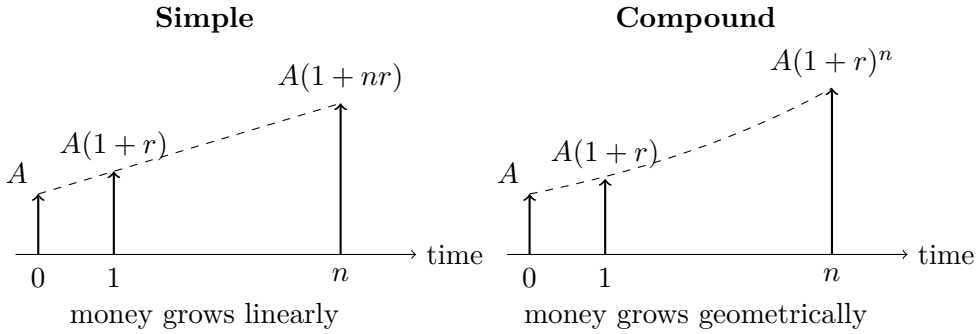


Ex. Invest \$1 for 1 year at 8%: after 1 year you receive

$$\$1.08 = \underbrace{\$1.00}_{\text{principal}} + \underbrace{\$0.08}_{\text{interest}}$$

Interest over multiple periods depend on the compounding rules:

1. **Simple**: Money grows linearly, receiving $V = A(1 + nr)$ over n periods.
2. **Compound**: Money grows geometrically (interest on interest), receiving $V = A(1+r)^n$ over n periods.



Rule of 72 (rule of thumb): money invested at r% per year doubles in approximately $72/r$ years.

Ex. *Simple:* \$5000 borrowed for 5 years at 10% per year simple interest is owed

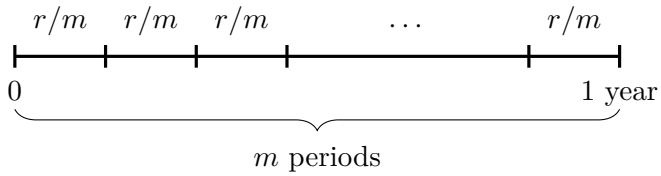
$$V = A(1 + rn) = 5000(1 + 0.1 \cdot 5) = \$7500$$

Compound: \$100 deposited at 7% compounded yearly for 10 years grows to

$$V = A(1 + r)^n = 100(1 + 0.07)^{10} = \$196.72$$

Simple interest is not used often in practice.

Def. **Nominal Interest Rate**: annual quote, compounding interest is in fractions corresponding to the compounding period.
For m periods per year for k periods, $V = A(1 + \frac{r}{m})^k$



Ex. Invest \$10 at 8% compounded quarterly for 3 quarters:

$$V = 10\left(1 + \frac{0.08}{4}\right)^3 = \$10.61$$

Def. **Continuous Compound Interest**: when the compounding period becomes arbitrarily small, the value approaches e^{rt}.

After 1 year with m periods: $(1 + \frac{r}{m})^m$, and $\lim_{m\to\infty}(1 + \frac{r}{m})^m = e^r$. More generally, in t years there are $k = tm$ periods, so $(1 + \frac{r}{m})^{tm} \to e^{rt}$.

Ex. $A = \$100$, $r = 9\%$ compounded continuously. After 1.5 years:

$$V = 100e^{(0.09)(1.5)} = \$114.45$$

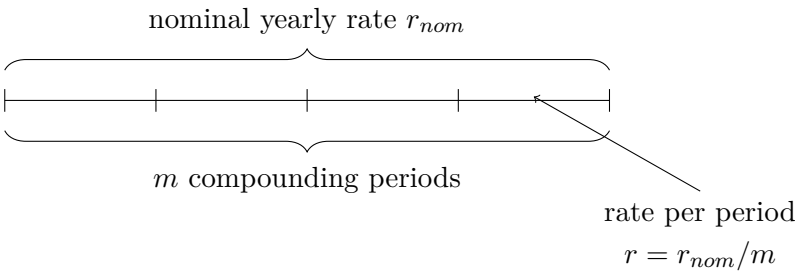
Compound Interest Summary

For r the nominal interest rate quoted:

Yearly for k years:	$(1+r)^k$
m periods per year for k periods:	$\left(1 + \frac{r}{m}\right)^k$
Continuously for time t:	e^{rt}

Def. **Effective Rate**: interest rates are usually quoted as nominal rates, but what you pay depends on the number of compounding periods. The equivalent rate with a *single* compounding period over the time interval of interest is the effective rate.

Relationship between Nominal and Effective Rate (with m compounding periods per year, so the rate per compounding period is $r = r_{nom}/m$):



$$1 + r_{eff} = (1 + \frac{r_{nom}}{m})^m \quad \Rightarrow \quad \boxed{r_{eff} = (1 + \frac{r_{nom}}{m})^m - 1}$$

1 + Simple One-year Rate = the Explicit Compound

Ex. Effective yearly rate of 8% compounded quarterly:

$$r_{eff} = \left(1 + \frac{0.08}{4}\right)^4 - 1 = 8.24\%$$

Def. **APR** (Annual Percentage Rate): the *nominal* annual rate of interest including all fees and charges.
APY (Annual Percentage Yield): the *effective* annual rate of interest including all fees and charges.

Both are required by the Truth in Lending Act (1968), and can be used to compare loans with different terms.

Ex. Credit Card
A credit card statement lists *daily* periodic rates of 0.02986% (purchases) and 0.05477% (advances).

	Nominal (APR)	Effective
Purchases	$(0.02986\%)(365) = 10.900\%$	$(1.0002986)^{365} - 1 = 11.514\%$
Advances	$(0.05477\%)(365) = 19.990\%$	$(1.0005477)^{365} - 1 = 22.121\%$

The statement's "annual percentage rate" is the nominal rate; the rate you actually pay is the higher effective rate.

Practice Problems

Ex. Yearly effective rate corresponding to:

(a) 8% compounded monthly: $\left(1+\frac{0.08}{12}\right)^{12}-1=8.30\%$

(b) 5% compounded semi-annually: $\left(1+\frac{0.05}{2}\right)^{2}-1=5.06\%$

(c) 10% compounded continuously: $e^{0.10}-1=10.52\%$

Ex. Convert by matching effective rates:

(a) 3% compounded quarterly to the nominal rate compounded monthly:

$$\left(1+\frac{r}{12}\right)^{12}=\left(1+\frac{0.03}{4}\right)^{4} \quad \Rightarrow \quad r=12\left[(1.0075)^{1/3}-1\right]=2.99\%$$

(b) 6% compounded semi-annually to the nominal rate compounded continuously:

$$e^{r}=\left(1+\frac{0.06}{2}\right)^{2} \quad \Rightarrow \quad r=2\ln(1.03)=5.91\%$$

Ex. Which rate would you choose? Compare the effective rates:

Quoted rate	Effective yearly rate
(a) 10% compounded continuously	10.52%
(b) 10.3% compounded monthly	10.80%
(c) 10.5% compounded quarterly	10.92%
(d) 10.6% compounded yearly	10.60%

As an investor, choose (c) since it has the highest effective rate (as a borrower, choose (a)).

Commonly Referred to Interest Rates

- **Discount Rate**: the rate at which member banks may borrow short term funds directly from a federal reserve bank.
- **Federal Funds Rate**: the rate that banks charge each other for the use of federal funds.
- **Prime Rate**: the interest rate that commercial banks charge their most creditworthy borrowers.
- **LIBOR** (London Inter-Bank Offer Rate): the interest rate that banks charge each other for loans (usually in Eurodollars).

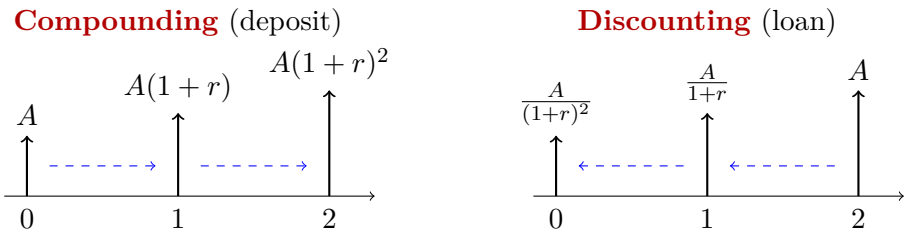
Lecture 2: Present Value and Internal Rate of Return

Sep 17, 2026

Present value: comparing cash flows that occur at different times, while interest rates tells how to move cash flows in time. Use interest rates to move all cash flows to the same time, so they become easy to compare.

Ideal Bank: applies the same rate of interest to deposits and loans, and has no transaction costs (real banks have higher rates for loans...). Used for two ways:

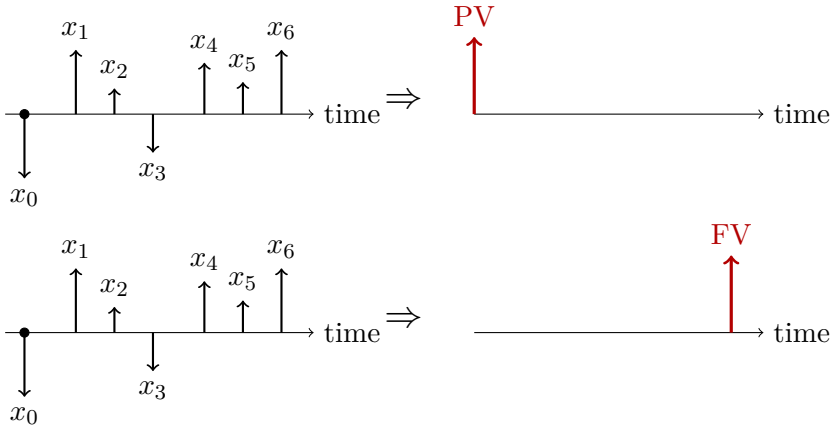
- **Compounding**: when cash flow is moved **forward**, **multiplied** by $(1+r)$ each period.
- **Discounting**: when cash flow is moved **backward**, **divided** by $(1+r)$ each period.



The Comparison Principle

The ideal bank brings cash flow stream to the same time, then is compared.

- When all cash flows are moved to *present time*, it is called **present value**.
- When all cash flows are moved to the *time of the final cash flow*, it is called **future value**.



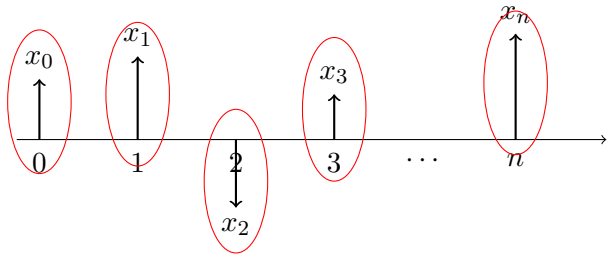
Ex. Cash flow stream $(-2, 1, 1, 1)$ at $r = 10\%$ (yearly compounding, but any compounding

convention works):

$$FV = -2(1.1)^3 + 1(1.1)^2 + 1(1.1) + 1 = 0.648$$
$$PV = -2 + \frac{1}{1.1} + \frac{1}{(1.1)^2} + \frac{1}{(1.1)^3} = 0.487$$

Check: $PV = FV/(1.1)^3 = 0.648/(1.1)^3 = 0.487$

General Formula for Present Value



Each cash flow (circled) is discounted separately, then all are summed. Let $x = (x_0, x_1, x_2, \ldots, x_n)$

$$PV(x) = x_0 + \frac{x_1}{(1+r)} + \frac{x_2}{(1+r)^2} + \cdots + \frac{x_n}{(1+r)^n}$$
$$= \boxed{\sum_{k=0}^{n} \frac{x_k}{(1+r)^k}}$$

Note that it's usually assumed that x_0 is negative, for the initial investment.

Main Theorem on Present Value

Two cash flow streams are equivalent for an ideal bank **if and only if** they have the same present value.

Importance: to compare two cash flow streams, compute their present values and compare.

Internal Rate of Return

Captures the idea on the "return" on an investment.
The return on your investment is the return which would make the present value equal to 0.

Ex. Invest \$100 today, receive \$110 in one year. What is your return?
Intuitively: 10%, why?

$$x = (-100, 110)$$

$$PV(x) = -100 + \frac{110}{1+0.1} = 0$$

Because under 10% interest, the present value of \$110 is exactly \$100.

Def. Let $x = (x_0, x_1, \ldots, x_n)$ be a cash flow stream.
Then, the **internal rate of return** of this stream is the number r satisfying:

$$PV(x) = 0$$
$$x_0 + \frac{x_1}{1+r} + \frac{x_2}{(1+r)^2} + \cdots + \frac{x_n}{(1+r)^n} = 0$$

To solve for this r, its more convenient to set

$$c = \frac{1}{(1+r)}$$

and solve the degree-n polynomial

$$x_0 + x_1 c + x_2 c^2 + \cdots + x_n c^n = 0$$

(by the fundamental theorem of algebra, it has n roots, possibly complex).

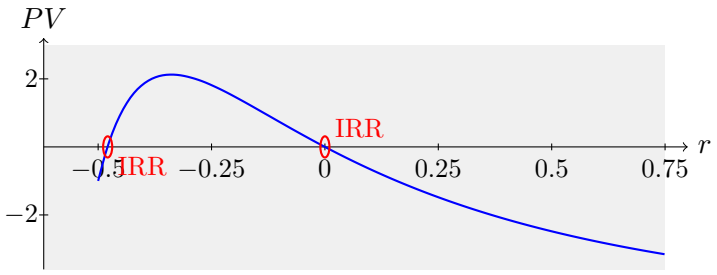
Ex. What is the internal rate of return of $(-2, 1, 1, 1)$?
Solve: $0 = -2 + c + c^2 + c^3$

$$c = 0.81 \quad \Rightarrow \quad c = \frac{1}{1+r} \quad \Rightarrow \quad r = 0.23$$

The polynomial can have many roots, which can be a problem. Fortunately, most standard investments have a well-defined IRR (see the theorem below).

Ex. Present Value Plot
Cash flow $(-5, -2, 12, -5)$: plotting PV against the interest rate r, the curve rises to a peak of about \$2.12 near $r = -0.35$, then falls and stays negative for all $r > 0$.

r	-0.5	-0.4	-0.3	-0.1	0	0.1	0.5
PV	-1.00	1.85	2.06	0.73	0.00	-0.66	-2.48



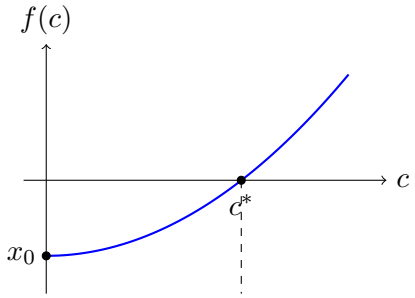
$PV = 0$ at both $r = 0$ and $r \approx -0.48$, so this stream has **two IRRs**. The sign of the cash flows changes more than once, so the Main Theorem on IRR below does not apply.

Main Theorem on IRR

Suppose the cash flow stream $x = (x_0, x_1, \ldots, x_n)$ has $x_0 < 0$ and $x_k \geq 0$ for all $k > 0$, where at least one term is strictly positive. Then, there is a unique positive root to the equation

$$x_0 + x_1 c + x_2 c^2 + \cdots + x_n c^n = 0$$

Proof: plot $f(c) = x_0 + x_1 c + x_2 c^2 + \cdots + x_n c^n$



Since $f(0) = x_0 < 0$ and $f'(c) = x_1 + 2x_2 c + \cdots + n x_n c^{n-1} \geq 0$ for $c \geq 0$ (as each $x_k \geq 0$, with at least one strictly positive), f is strictly increasing for $c > 0$ and unbounded above. So f crosses zero exactly once, at some $c^* > 0$, giving the unique IRR $r = 1/c^* - 1 > -1$.

Net Present Value (NPV)

Computes all present values of *all* cash flows associated with an investment. Includes the investment term x_0 as well.

Ex. When to Cut a Tree

You can plant trees to be sold for lumber later, with two options for when to cut (interest rate 10%):

(1) After 1 year: $(-1, 2)$
(2) After 2 years: $(-1, 0, 3)$

Using NPV (choose largest NPV):

$$-1 + \frac{2}{1.1} = 0.82 \quad \text{vs} \quad -1 + \frac{0}{1.1} + \frac{3}{1.1^2} = 1.48$$

NPV says choose option (2).
Using IRR (choose greatest IRR):

$$-1 + 2c = 0 \Rightarrow c = 0.5 \Rightarrow r = 1.0 \quad \text{vs} \quad -1 + 3c^2 = 0 \Rightarrow c = 0.577 \Rightarrow r = 0.7$$

IRR says choose option (1).

IRR and NPV give conflicting recommendations.

IRR is scale invariant, Present Value is not.
PV distinguishes between single and repeated cash flows, IRR does not.

In general, NPV is preferred.

Ex. Scaling and repetition (PV at 10%):

Cash flows	$(-1, 2)$	$10 \cdot (-1, 2)$	$(-1, 2, -1, 2, \dots)$
PV	0.818	8.18	4.71
IRR	$r = 1.0$	$r = 1.0$	$r = 1.0$

NPV is preferred because it distinguishes between more cash flows, but this means the cash flows and assumptions need to be modelled carefully before the NPV analysis is done. E.g. the tree-cutting NPV analysis assumes it is done only *once*; if the process can be repeated many times, we arrive at a different answer. Methods of IRR analysis exist that make it agree with NPV (e.g. incremental IRR analysis).

Steps to Solving Cash Flow Problems

1. Write down the cash flow diagram for each alternative (include *all* cash flows).
2. Evaluate the NPV of each cash flow.
3. Choose the alternative with the largest NPV.

Cycle Problems

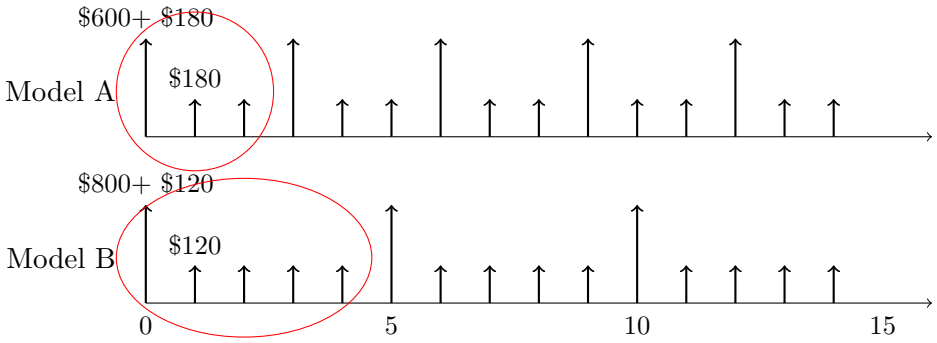
Relate to activities that can be repeated. With NPV you arrive at a different answer depending on how many times an activity is repeated, so the activities must be evaluated over the same time horizon. Two approaches:

1. Repeat both activities until they terminate at the same time.
2. Repeat both to infinity: for an activity with lifetime T and one-cycle present value PV_1,

$$PV_1 \sum_{k=0}^{\infty} \frac{1}{(1+r)^{Tk}} = PV_1 \cdot \frac{1}{1 - \frac{1}{(1+r)^T}}$$

Ex. Refrigerator Purchase

Model A: price \$600, yearly cost \$180 (beginning of year), life 3 years.
Model B: price \$800, yearly cost \$120 (beginning of year), life 5 years.
Interest rate is 10%. Over 15 years (costs drawn as arrows, tall arrows mark a new purchase):



One cycle of each (circled; price and first cost at time 0):

$$PV_A = 780 + 180 \sum_{k=1}^{2} \frac{1}{1.1^k} = \$1,092.40 \qquad PV_B = 920 + 120 \sum_{k=1}^{4} \frac{1}{1.1^k} = \$1,300.38$$

Approach 1: equal horizons, 15 years = 5 cycles of A = 3 cycles of B

$$A : PV_A \left(1 + \frac{1}{1.1^3} + \frac{1}{1.1^6} + \frac{1}{1.1^9} + \frac{1}{1.1^{12}}\right) = \$3,341.11$$
$$B : PV_B \left(1 + \frac{1}{1.1^5} + \frac{1}{1.1^{10}}\right) = \$2,609.17$$

Approach 2: repeat to infinity

$$A : 1092.40 \cdot \frac{1}{1 - \frac{1}{1.1^3}} = \$4,392.70$$
$$B : 1300.38 \cdot \frac{1}{1 - \frac{1}{1.1^5}} = \$3,430.37$$

These are costs, so the smaller is better: choose Model B in both approaches.

Summing Geometric Series

$$S = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots$$
$$= 1 + x(1 + x + x^2 + \cdots) = 1 + xS$$

$$S(1 - x) = 1 \quad \Rightarrow \quad S = \frac{1}{1 - x} \qquad (|x| < 1)$$

Lecture 3: Fixed Income Securities

Sep 21 & 24, 2026

Def. Financial Instruments

A legal obligation or claim having monetary value (eg. mortgage, stocks and bonds...)

Def. Security

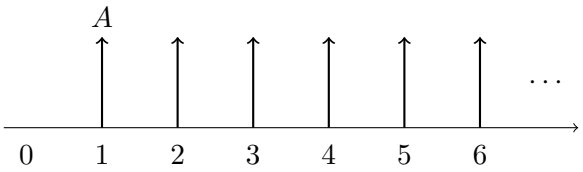
A tradable financial instrument satisfying legal and regulatory requirements.

Def. Fixed Income Securities

Securities that promise a fixed income to the holder over some span of time. (eg. bonds, mortgages, annuities)

Annuity

A contract that pays the holder money periodically according to the fixed schedule.
Perpetual Annuity (Perpetuity) - Pays a fixed sum periodically, forever.



At a per period interest rate of r, the PV is

$$P = \sum_{k=1}^{\infty} \frac{A}{(1+r)^k} = \frac{A}{r}$$

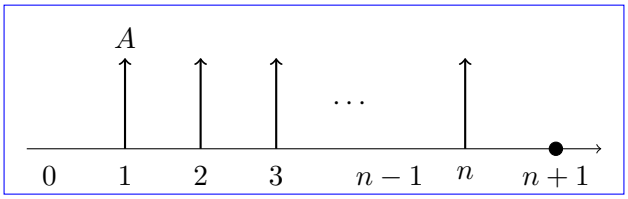
Note: Infinite sum!!

Ex. A perpetual annuity of \$1,000 each year at 10% interest/year has present value

$$P = \frac{A}{r} = \frac{\$1,000}{0.1} = \$10,000$$

Finite-Life Streams

A stream that pays A for n periods, starting at period 1. What is the PV?

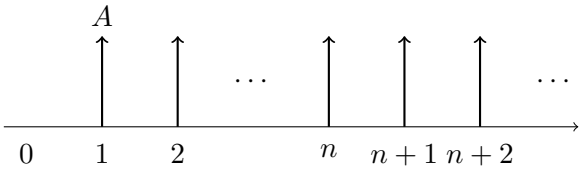


$$P = \sum_{k=1}^{n} \frac{A}{(1+r)^k} = \frac{A}{r}\left(1 - \frac{1}{(1+r)^n}\right)$$

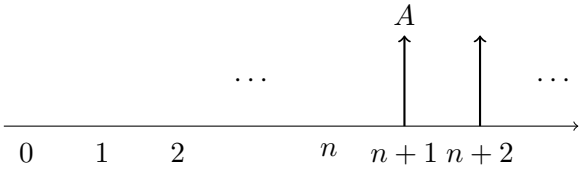
where r is the interest rate per period.

A finite life-stream can be thought as the difference between two perpetuities.

Perpetuity starting at period 1:



Perpetuity starting at period $n+1$:



$$PV = \frac{A}{r} - \frac{A}{r}\left(\frac{1}{(1+r)^n}\right)$$

$$PV_{fin} = \frac{A}{r}\left(1 - \frac{1}{(1+r)^n}\right)$$

Ex. Loan Calculation

Suppose \$1,000 is borrowed at 12% interest compounded monthly, repaid with equal monthly payments over 5 years. How much are the monthly payments?

Given: $P = \$1,000 \quad r = \frac{12\%}{12} = 1\%/\text{mo} \quad n = 5 \times 12 = 60$

$$P = \frac{A}{r}\left(1 - \frac{1}{(1+r)^n}\right) \quad \Rightarrow \quad A = \frac{rP}{\left(1 - \frac{1}{(1+r)^n}\right)} = \$22.24/\text{mo}$$

Bonds

An agreement to pay money according to the rules of the issue.
Generally, bonds have

- Face Value, Par Value, Principal - an amount to be paid at the maturity date.

- Coupon Payments - an amount paid periodically, expressed as a % of the face value. Last coupon is paid at maturity.

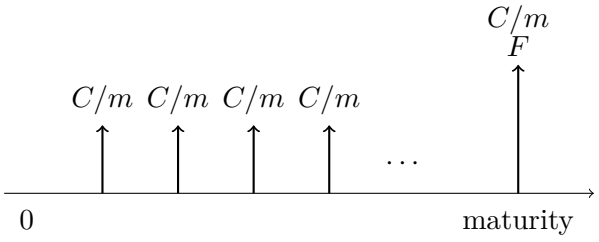
Ex. Given:

$$\text{Face Value } = \$1,000 \qquad \text{Coupon } = 9\%$$

The yearly coupon amount is $\$1,000(0.09) = \$90$

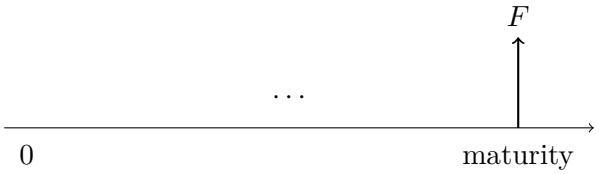
In general, coupons are paid m times per year, so each payment is C/m where C is the yearly coupon amount. In the U.S., coupons are usually paid every six months ($m = 2$), e.g. US treasury bonds with this example pay \$45 every 6 months (\$90/2).

Payoff Structure of a Bond



where C is the annual coupon amount, and F is the face value.

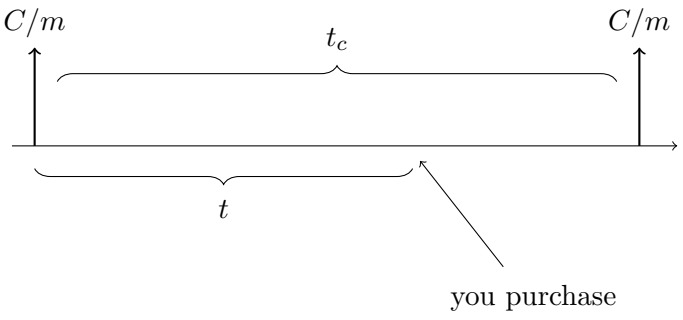
When there are no coupons, the only payoff is the face value at maturity.



This is known as a **zero-coupon bond**, or a zero.

Accrued Interest

Ignored by bond price quotes, but must be added to the price.



You must pay the previous owner their "portion" of the next coupon payment.

$$AI = \frac{\text{days since last coupon}}{\text{days in current coupon period}} \times (C/m) = \frac{t}{t_c}(C/m)$$

Ex. A treasury bond is purchased on May 8 and matures on August 15 in some future year. The coupon rate is 9% and coupons are paid every Feb 15 and Aug 15. How much accrued interest is owed?
On May 8th, 83 days have passed since the last coupon and 99 days remain until the next. With $C/m = 9\%/2 = 4.5\%$:

$$AI = \frac{83}{83+99} \times 4.5\% = 2.05\%$$

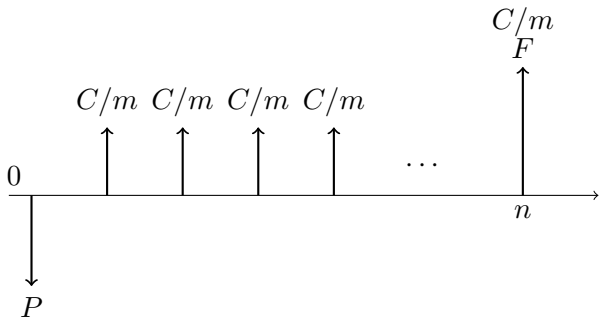
Def. **Clean price**: the market bond price (quoted, doesn't include accrued interest).
Dirty (cash) price: the clean price plus accrued interest, i.e. what you actually pay.

Government Securities:

- Bill: up to 1 year
- Note: 1 to 10 years
- Bond: 10 to 30 years

Bond Price and Yield

The whole cash flow stream of purchasing a bond can be mapped out, where P is the entire amount paid for the bond: the market price plus accrued interest (the dirty price).



The IRR of this cash flow stream is the **yield** (or **yield to maturity**) of the bond, denoted λ. It captures the rate of return from buying the bond and holding it to maturity.

$$P = \underbrace{\frac{F}{\left(1+\frac{\lambda}{m}\right)^n}}_{\text{PV of face}} + \underbrace{\frac{C/m}{\lambda/m}\left[1 - \frac{1}{\left(1+\frac{\lambda}{m}\right)^n}\right]}_{\text{annuity formula}}$$

P : price including accrued interest
F : face value
m : coupon payments per year (compounding m times per year)
C/m : amount of each coupon payment
λ/m : yield per coupon period
n : coupon payments remaining

This formula relates price and yield only with **exactly** n coupon payments remaining (i.e. right after a coupon is paid).

Ex. Price of a bond with par value \$1,000 paid in ten years, coupon rate 10%, yield 12%, coupons paid semi-annually. Then $n = 20$, $C/m = 50$, $\lambda/m = 6\%$:

$$P = 50 \cdot \frac{1 - \frac{1}{(1.06)^{20}}}{0.06} + \frac{1000}{(1.06)^{20}} = 50(11.47) + 311.80 \approx \$885.30$$

Def. Comparing a bond's price P to its face value F (equivalently, its coupon rate C/F to its yield λ):

- **Par bond**: $P = F$, when the coupon rate **equals** the yield.
- **Premium bond**: $P > F$, when the coupon rate is **above** the yield.
- **Discount bond**: $P < F$, when the coupon rate is **below** the yield.

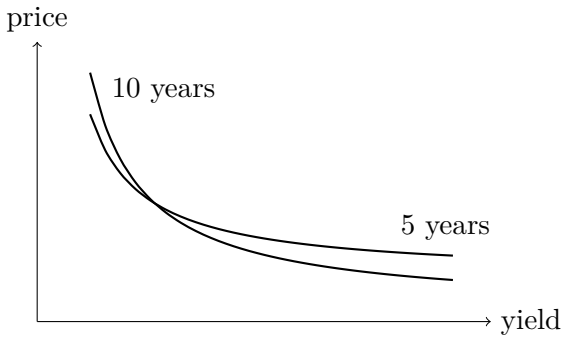
E.g. the bond above has a 10% coupon but a 12% yield, so it sells at a discount ($\$885.30 < \$1,000$).

The Price-Yield Curve

For a fixed maturity, price and yield are inversely related:

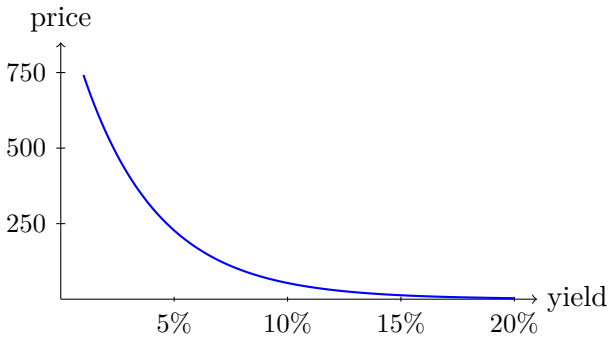
- Rates Rise → Existing bond prices Fall
- Rates Fall → Existing bond prices Rise

The longer the time to maturity, the more sensitive the price of the bond is to the yield.



Ex. A 30-year zero-coupon bond with face value \$1,000, priced at different yields (compounded semi-annually):

Yield	1%	2%	5%	10%	15%	20%
Price	\$741.37	\$550.45	\$227.28	\$53.54	\$13.05	\$3.28



Price falls steeply as yield rises: $P = 1000/\left(1+\frac{\lambda}{2}\right)^{60}$.

The **current yield** is the coupon divided by the current price. In this course, **yield** always means **yield to maturity**.

Ex. Bond Quote
Treasury quotes from yahoo.com (YTM = yield to maturity, CY = current yield):

Issue	Price	Coupon	YTM	CY
T-Note, matures 31-Oct-2005	99.91	1.625	2.923	1.626
T-Note, matures 15-Jan-2011	109.71	3.500	1.574	3.190
T-Bond, matures 15-Feb-2021	135.49	7.875	4.622	5.812

Check: $CY = 7.875/135.49 = 5.812\%$. YTM and CY can differ a lot, since CY ignores the face value repaid at maturity.

Bond Ratings

Standard & Poor's and Moody's are two independent bond rating agencies.

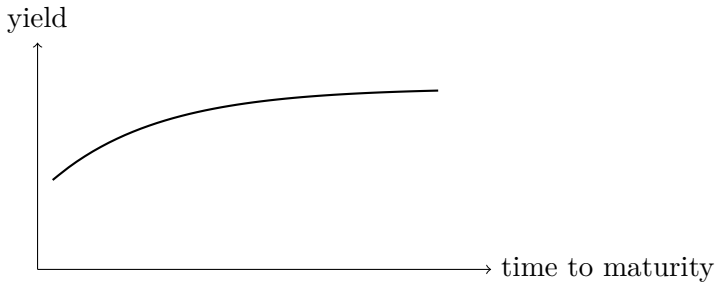
	S&P	Moody's	Meaning
Investment	AAA	Aaa	Best quality, smallest credit risk (US gov. bonds)
Grade	AA	Aa	High grade
	A	A	High to medium grade
	BBB	Baa	Medium grade
Speculative	BB	Ba	Judged to be speculative
Grade	B	B	Increasingly speculative
(Junk Bonds)	CCC	Caa	Danger of default
	CC	Ca	Very speculative, high chance of default
	C	C	Small chance of not defaulting
	D	D	In default

Lecture 4: The Term Structure of Interest Rates

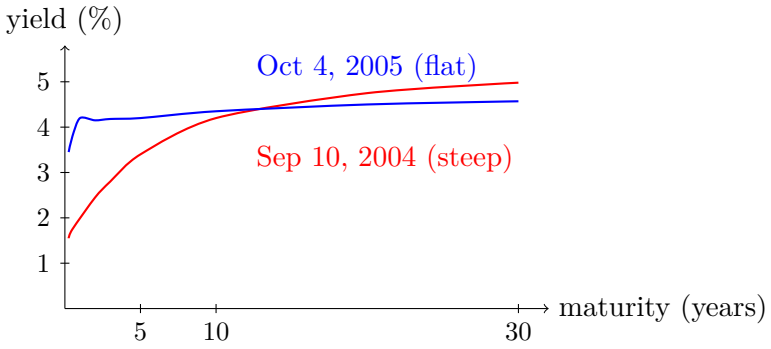
Sep 28, 2026

The Yield Curve

In general, yields on bonds of different maturities are different, i.e. interest rates are a function of time.
Plotting yield vs. time to maturity for bonds gives the **yield curve**.



Real U.S. Treasury yield curves (yahoo.com): on Sep 10, 2004 the curve rose steeply from about 1.5% (3 months) to 5% (30 years); on Oct 4, 2005 it was much flatter, from about 3.5% to 4.6%. The shape of the curve changes over time.



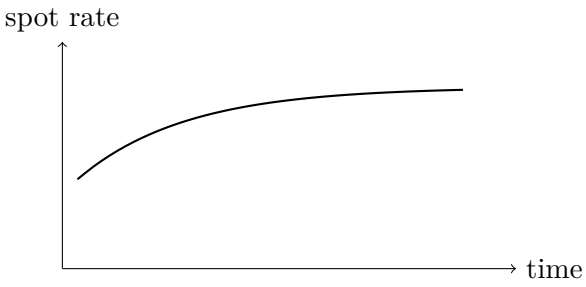
The yield curve is not exactly what we want, since bonds may have coupons (intermediate cash flows). We would really like to know the interest rate on a **single cash flow at time** t, with no intermediate cash flows.



Spot Rates

Def. Spot Rate

Interest rates for a specific time t. Always **annual nominal interest rates**, can be compounded yearly, m periods per year, or continuously.



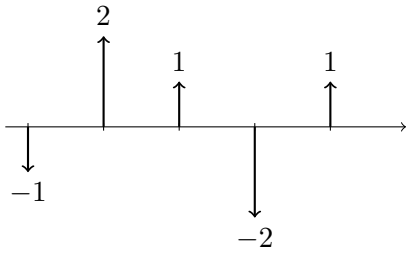
Plotting spot rates vs. time gives the **spot rate curve**, also known as the **term structure of interest rates**.

We can price bonds reliably with predictions through the term structure of interest (quant!!).

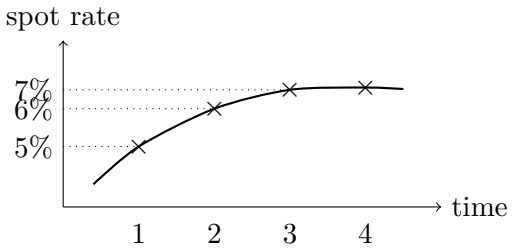
Present Value and Spot Rates

When computing present value, the spot rate corresponding to the time of each cash flow must be used for discounting.

$$PV = x_0 + \frac{x_1}{1+s_1} + \frac{x_2}{(1+s_2)^2} + \cdots + \frac{x_n}{(1+s_n)^n}$$



Ex.



Time	0	1	2	3	4
Cash flow	-1	2	1	-2	1
Spot rate		5%	6%	7%	7%

$$PV = -1 + \frac{2}{1.05} + \frac{1}{1.06^2} + \frac{-2}{1.07^3} + \frac{1}{1.07^4} = 0.925$$

Determining Spot Rates

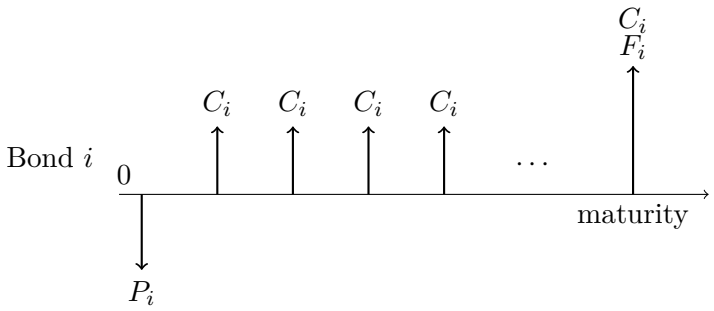
If you know the price of a **zero-coupon bond**, its yield is the spot rate. Given price P and face value F of a zero maturing at t:

$$\boxed{P = Fe^{-s_t t}} \quad \text{(continuous compounding)} \qquad \boxed{P = \frac{F}{(1+s_t)^t}} \quad \text{(discrete compounding)}$$

Problem: What if there are no zeros?
Solution: Construct a zero-coupon bond from coupon bonds.

Constructing Zeros

Take two coupon bonds with the same maturity (and coupon dates) and form a portfolio of x of bond 1 and y of bond 2 (negative means short).



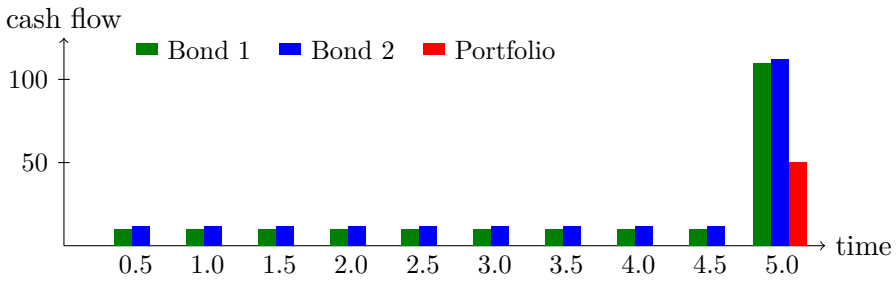
Three equations for the portfolio:

$$\begin{aligned} \text{Coupon:} \quad & xC_1 + yC_2 = C_0 \\ \text{Face:} \quad & xF_1 + yF_2 = F_0 \\ \text{Price:} \quad & xP_1 + yP_2 = P_0 \end{aligned}$$

To construct a zero, set $C_0 = 0$ and $F_0 = \$100$ (e.g. with $F_1 = F_2 = \$100$). Then solve for x and y. The price of the zero is then $P_0 = xP_1 + yP_2$, and the spot rate could be computed.

Ex. Portfolio of 3 of bond 1 (face \$100, 10% coupon) and -2.5 of bond 2 (face \$100, 12% coupon), both maturing at $t = 5$:

Time	0.5	1	$\cdots$	5
Bond 1	10	10	$\cdots$	110
Bond 2	12	12	$\cdots$	112
Portfolio	0	0	$\cdots$	50



Coupons cancel: $3(10) - 2.5(12) = 0$, leaving only $3(110) - 2.5(112) = 50$ at maturity, a zero.

Ex. Bond 1: 10 year, 10% coupon, $P_1 = \$98.72$. Bond 2: 10 year, 8% coupon, $P_2 = \$85.89$. (Note: Choose two bonds with the **same maturity**, so the spot rate for that maturity can be computed)
Create a zero:

$$\begin{cases} \text{Coupon:} & 10x + 8y = 0 \\ \text{Face:} & 100x + 100y = 100 \end{cases} \quad \Rightarrow \quad x = -4,\ y = 5$$

Price of zero: $xP_1 + yP_2 = -4(98.72) + 5(85.89) = \$34.57$
Spot Rate (discrete compounding):

$$34.57 = \frac{100}{(1 + s_{10})^{10}} \quad \Rightarrow \quad s_{10} = 0.112 = 11.2\%$$

Note: if possible, you can compare this computed value with the actual yield, then see which one is "better"

Forward Rates

Def. **Forward Rates**
The interest rate for money to be borrowed between two dates in the future, but under terms agreed

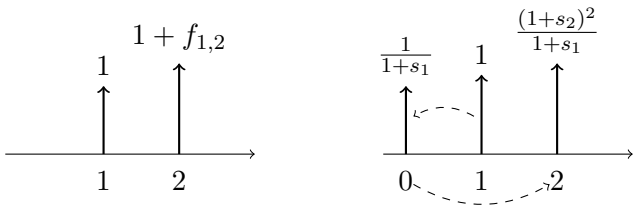
upon today.

$$f_{t_1,t_2} : \text{the forward rate between } t_1 \text{ and } t_2.$$

Note: t_1 can be 0, but not negative, when $t_1 = 0$, $f_{0,t_2} = s_{t_2}$, the spot rate of t_2.

How can forward rates be computed from the spot rate curve?

Starting with \$1 at time 1 and moving it to time 2, we should have $1 + f_{1,2}$ by the definition of the forward rate. Spot rates tell us how to move money to and from time 0, so we can instead go through time 0:



$$1 + f_{1,2} = \frac{(1+s_2)^2}{1+s_1} \quad \Rightarrow \quad \boxed{f_{1,2} = \frac{(1+s_2)^2}{1+s_1} - 1}$$

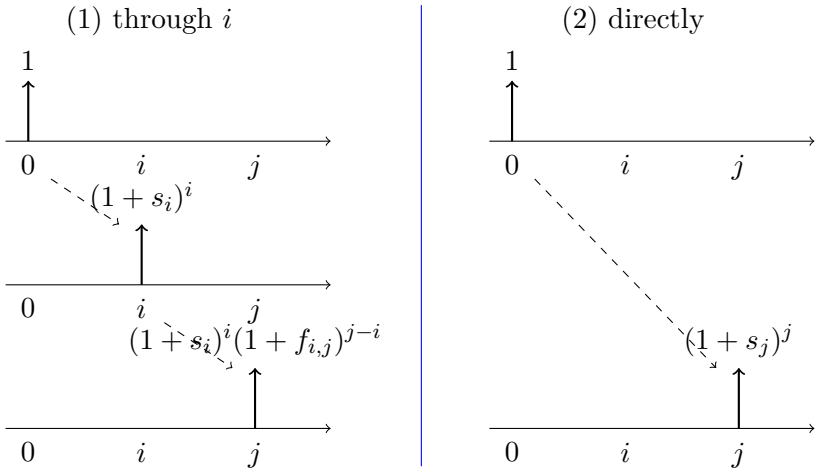
Ex. Spot rates for 1 and 2 years are 7% and 8%. The implied forward rate is

$$f_{1,2} = \frac{(1.08)^2}{1.07} - 1 = 0.0901$$

Note that the forward rate has to be **larger** than 8%: the first year only earns 7%, so the second year must make up the difference to reach $(1 + 0.08)^2$.

Formula for Forward Rates

Invest \$1 at time 0 until time j in two ways: (1) at the spot rate s_i until i, then at the forward rate $f_{i,j}$ until j; (2) directly at the spot rate s_j. Both must give the same result:



$$\boxed{(1+s_i)^i(1+f_{i,j})^{j-i} = (1+s_j)^j} \qquad f_{i,j} = \left[\frac{(1+s_j)^j}{(1+s_i)^i}\right]^{1/(j-i)} - 1$$

Note: this calculation must be consistent with the type of compounding.

Ex. Spot rates for 2 and 6 years are 3% and 5%. The implied forward rate is

$$(1+s_2)^2(1+f_{2,6})^4 = (1+s_6)^6 \quad \Rightarrow \quad f_{2,6} = \left(\frac{(1.05)^6}{(1.03)^2}\right)^{1/4} - 1 = 6.01\%$$

Def. Rates that span only 1 period are **short rates**. E.g. for yearly compounding: $s_1, f_{1,2}, f_{2,3}, \ldots$

Short rates can be determined from the spot rate curve and vice versa.

Term Structure Explanations

Why isn't the term structure (spot rate curve) flat? Three explanations:

(1) **Expectations Theory**: spot rates are determined by the expectations of what interest rates will be. E.g. if the spot rate curve slopes upward, it is because people expect interest rates to increase in the future.

(2) **Liquidity Preference**: investors prefer short term securities over long term ones, since long term securities "tie up" capital that may be needed before maturity. However, there are well developed markets for both short and long term securities, so investors can easily get out. More likely, long term investments are **more sensitive to interest rates**, so to lessen risk investors prefer short term securities (and demand higher rates for long term ones).

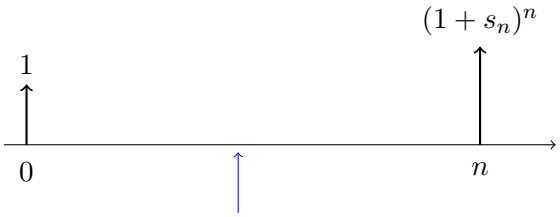
(3) **Market Segmentation**: the fixed income market is segmented by maturity dates. The rate for each maturity is determined by the group of investors interested in that maturity, i.e. simple supply and demand among those investors.

Expectation Dynamics

Forward rates are rates in the future that are agreed upon **now**. But we don't know what the **actual future rates** will be when that time arrives, since spot rate curves could change. When forward rates are used as a prediction (could be wrong) of the actual future rates, this forecast is called **expectation dynamics**.
Prohibits arbitrage, using forward rates that are anchored on the spot rates today.

The Invariance Theorem

If interest rates evolve according to expectation dynamics, then a sum of money invested in the interest rate market for n years, will grow by a factor of $(1+s_n)^n$ regardless of the investment and reinvestment strategy (no matter what combination of spot and forward rates).



Why? Implied forward rates were computed to satisfy exactly this relationship.
So even if you don't lock in a future interest rate now, you can still use forward rates for discounting, as long as you assume expectation dynamics.

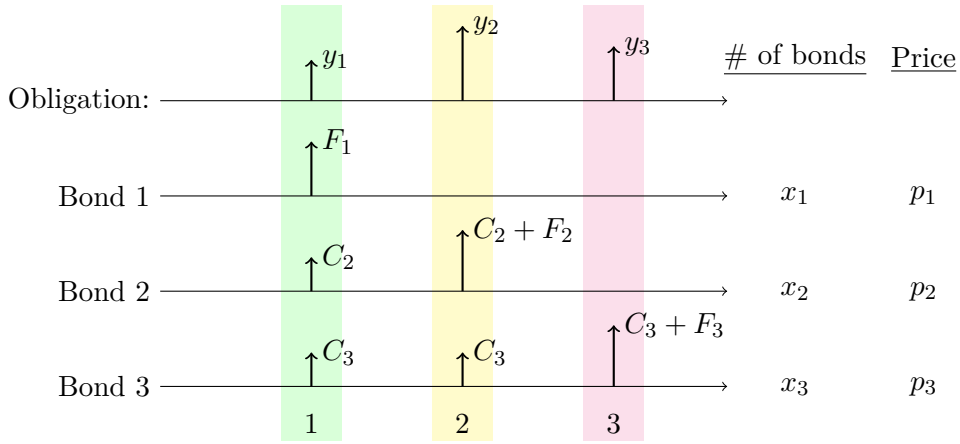
If expectation dynamics are assumed, the invariance theorem is true (will always equal the same return value regardless of investment strategies.)

Lecture 5: Cash Matching

Oct 1, 2026

Cash Matching Problem

A series of future obligations must be met.
You have various bonds you can invest in, where the portfolio could be used to meet the obligation.
Choose portfolio of minimum cost that has enough money to meet the obligation (optimization!!)



Let y_t be the obligation at time t, and x_i the number of bond i (price p_i, coupon C_i, face F_i) we buy.

Assumptions (the problem can be solved under each one):

1) No reinvestment of excess cash is allowed.
2) Excess cash can be held at zero interest from one period to the next.
3) Excess cash can be reinvested at the appropriate forward rate.

Constrained optimization?????
Yes: the objective (total cost) and every constraint are **linear** in the unknowns, so each model is a **linear program** (LP), solvable with e.g. Excel Solver.

Note # of bonds should always be non-negative! ($x_i \geq 0$, no short selling)

Model 1: Without Reinvestment

Each period's cash flow from the bonds must cover that period's obligation on its own:

$$\begin{aligned} \min_{x_1,x_2,x_3} \quad & p_1x_1 + p_2x_2 + p_3x_3 \\ \text{s.t.} \quad & F_1x_1 + C_2x_2 + C_3x_3 \geq y_1 \\ & (F_2 + C_2)x_2 + C_3x_3 \geq y_2 \\ & (F_3 + C_3)x_3 \geq y_3 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

Model 2: Reinvestment at Zero Rate

z_i is the possible excess cash generated, when positive, could be carried forward for reinvestment. Excess z_1 is subtracted at time 1 and added back (with no interest) at time 2:

$$\begin{aligned} \min_{x_1,x_2,x_3} \quad & p_1x_1 + p_2x_2 + p_3x_3 \\ \text{s.t.} \quad & F_1x_1 + C_2x_2 + C_3x_3 - z_1 \geq y_1 \\ & (F_2 + C_2)x_2 + C_3x_3 + z_1 - z_2 \geq y_2 \\ & (F_3 + C_3)x_3 + z_2 \geq y_3 \\ & x_i \geq 0, \quad z_1, z_2 \geq 0 \end{aligned}$$

Model 2 will never be worse than model one, worst case the same ($z = 0$) than model 1, thus it is superior.

Model 3: Reinvestment at Forward Rate

Model 3: investment with forward rate. Excess z_1 carried from time 1 to 2 now grows by $(1 + f_{1,2})$:

$$\begin{aligned} \min_{x_1,x_2,x_3} \quad & p_1x_1 + p_2x_2 + p_3x_3 \\ \text{s.t.} \quad & F_1x_1 + C_2x_2 + C_3x_3 - z_1 \geq y_1 \\ & (F_2 + C_2)x_2 + C_3x_3 + (1 + f_{1,2})z_1 - z_2 \geq y_2 \\ & (F_3 + C_3)x_3 + (1 + f_{2,3})z_2 \geq y_3 \\ & x_i \geq 0, \quad z_1, z_2 \geq 0 \end{aligned}$$

Similarly, as long as forward rates are non-negative, $(1 + f)z \geq z$, so Model 3 is never worse than Model 2.

Ex. Cash Matching Spreadsheet

Four bonds (face \$100, annual coupons), bond i maturing at time i, and obligations $(10, 200, 30, 50)$ at times 1–4.

The term structure uses **continuous compounding**: discount factors $d(0, i) = e^{-s_i i}$, and reinvesting from $i - 1$ to i multiplies by $e^{f_{i-1,i}}$ (e.g. $e^{0.04} = 1.0408$).

	Term structure			Bond cash flows				
Time	s_i	$f_{i-1,i}$	$d(0,i)$	1 (3%)	2 (4%)	3 (2%)	4 (5%)	Obligation
1	3%	3.0%	0.9704	103	4	2	5	10
2	3.5%	4.0%	0.9324		104	2	5	200
3	4%	5.0%	0.8869			102	5	30
4	4.2%	4.8%	0.8454				105	50
			Price	99.96	100.85	94.27	102.71	

Each price is the bond's cash flows discounted at the spot rates, e.g. $p_1 = 103(0.9704) = 99.96$. Optimal portfolios:

Model	x_1	x_2	x_3	x_4	Total cost
1) No reinvestment	0	1.8950	0.2708	0.4762	\$265.546
2) Zero rate ($z_1 = 0.484$)	0	1.8903	0.2708	0.4762	\$265.077
3) Forward rate ($z = 21.25, 9.97, 9.19$)	0.2104	1.7828	0.2626	0.3843	\$265.059

- In Model 1, time 1 receives \$10.50 > \$10: the extra \$0.50 is wasted, since it can't be carried forward.
- In Model 2, that excess ($z_1 = 0.484$) is carried to time 2, so slightly fewer of bond 2 are needed.
- Cost decreases $1 \to 2 \to 3$, as expected.
- Model 3's cost equals the PV of the obligations at the spot rates: $10(0.9704) + 200(0.9324) + 30(0.8869) + 50(0.8454) =$ \$265.059. With reinvestment at the forward rates, every dollar moves through time exactly as the term structure says, so no portfolio can do better than this.

Ex. Writing Out the Linear Programs

Same four bonds and obligations as above, with x_j = units of bond j and z_t = cash carried from year t to year $t+1$. All three models share the objective (prices from the term structure):

$$\min \quad 99.955890x_1 + 100.850739x_2 + 94.271563x_3 + 102.710952x_4$$

1) No reinvestment:

$$\begin{aligned} 103x_1 + 4x_2 + 2x_3 + 5x_4 &\geq 10 \\ 104x_2 + 2x_3 + 5x_4 &\geq 200 \\ 102x_3 + 5x_4 &\geq 30 \\ 105x_4 &\geq 50 \\ x_1, x_2, x_3, x_4 &\geq 0 \end{aligned}$$

$x^* = (0, 1.894976, 0.270775, 0.476190)$, minimum cost \$265.546082. Year 1 has an excess of 0.502406 that is lost.

2) Carry cash at zero interest: each z_t leaves year t and arrives unchanged in year $t+1$.

$$\begin{aligned} 103x_1 + 4x_2 + 2x_3 + 5x_4 - z_1 &\geq 10 \\ 104x_2 + 2x_3 + 5x_4 + z_1 - z_2 &\geq 200 \\ 102x_3 + 5x_4 + z_2 - z_3 &\geq 30 \\ 105x_4 + z_3 &\geq 50 \\ x \geq 0, \quad z_1, z_2, z_3 &\geq 0 \end{aligned}$$

$x^* = (0, 1.890324, 0.270775, 0.476190)$, $z^* = (0.483798, 0, 0)$, minimum cost \$265.076934. Cash left after year 1 is available in year 2.

3) Reinvest at the forward rate: each z_t grows by the forward factor $e^{f_{t,t+1}}$ before arriving, i.e. $e^{0.04}$, $e^{0.05}$, $e^{0.048}$.

$$\begin{aligned} 103x_1 + 4x_2 + 2x_3 + 5x_4 - z_1 &\geq 10 \\ 104x_2 + 2x_3 + 5x_4 + 1.040811z_1 - z_2 &\geq 200 \\ 102x_3 + 5x_4 + 1.051271z_2 - z_3 &\geq 30 \\ 105x_4 + 1.049171z_3 &\geq 50 \\ x \geq 0, \quad z_1, z_2, z_3 &\geq 0 \end{aligned}$$

Excel's portfolio, with minimum cost \$265.058524:

$$x^* \approx (0.210372,\ 1.782777,\ 0.262646,\ 0.384342), \qquad z^* \approx (21.24643,\ 9.96928,\ 9.192063)$$

Model	Minimum initial cost
No reinvestment	\$265.546082
Zero interest carry	\$265.076934
Forward rate carry	\$265.058524

Allowing cash to move across years reduces the purchase cost.

At forward rates, **several portfolios** have the same minimum cost, so the optimal x^* is not unique (Excel's answer is just one of them). This matches the note above: any portfolio that exactly meets the obligations costs the PV of the obligations.

Lecture 6: Bond Price Sensitivity

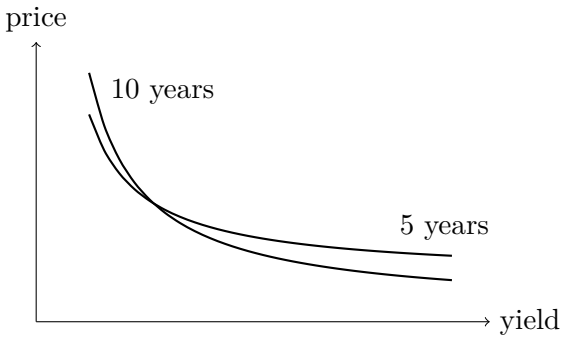
Oct 5, 2026

Bond price sensitivity is measured two ways:

- **Yield sensitivity**: Macaulay duration, modified duration, and the connection between them.
- **Spot rate curve sensitivity**: Fisher-Weil duration and quasi-modified duration.

Question: I would like to invest in a bond, but I am worried that bond yields may change in the future. Should I invest in a short or long maturity bond to reduce the effect of yield changes on the value of my bond? Should I choose a bond with large or small coupons?
Answer: We need a way to compute the **sensitivity of bond prices to yield/interest**.

Sensitivity Related to Time to Maturity



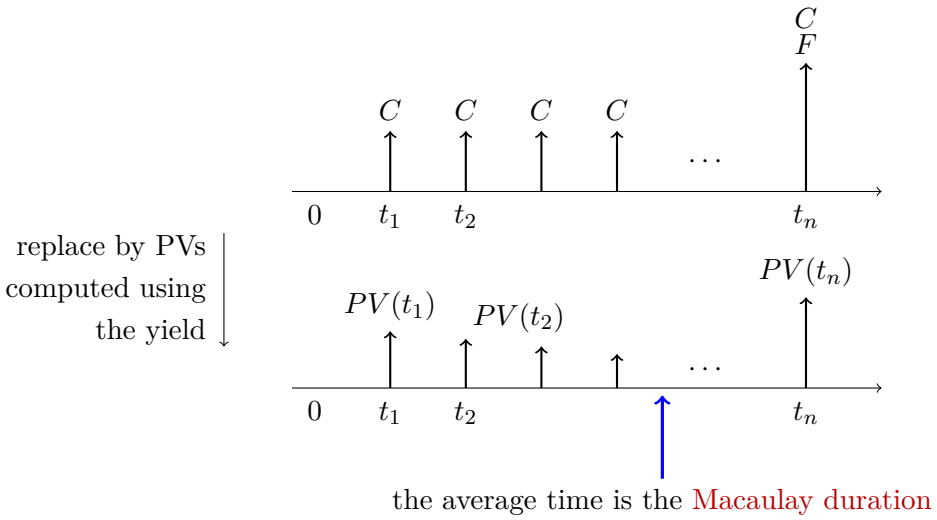
The longer the time to maturity, the more sensitive the price of the bond is to the yield.

Duration

Maturity time is related to the sensitivity of the bond price to its yield.
In the case of zero-coupon bonds, the **duration** (numerical measure for sensitivity) is the maturity (time measure), as the center of gravity is completely on the singular cash flow at maturity.
When there are coupons, the relationship is no longer direct.

Macaulay Duration

Take the **weighted average** of the times to the cash flows, where each cash flow is weighted by its **present value** (computed with the yield) divided by the total present value.



Def. **Macaulay Duration**

$$D = \frac{PV(t_1)t_1 + PV(t_2)t_2 + \cdots + PV(t_n)t_n}{PV_{total}}$$
$$= w_{t_1}t_1 + w_{t_2}t_2 + \cdots + w_{t_n}t_n$$
$$\text{where } w_{t_i} = \frac{PV(t_i)}{PV_{total}}$$

Where $PV(t_i)$ is the present value of the cash flow at time t_i, computed using the yield as the interest rate, and $PV_{total} = \sum_i PV(t_i)$ is the present value of the entire cash flow.
Macaulay duration is a **weighted average of times**, so it is quoted in **years**.

Economic meaning: imagine buying a bond today. Macaulay duration answers "if I view all future payments as one equivalent payment, at what average future date would I receive it?" It is an **average waiting time to receiving cash**.

Macaulay Duration and Maturity

- Zero-coupon bond: Macaulay duration = maturity date.
- Coupon bond: Macaulay duration < maturity date.

You should think: a bond with **Macaulay duration *D*** has the same yield sensitivity as a **zero-coupon bond with maturity *D***.

Back to the opening question: to reduce the effect of yield changes, choose a bond with a **shorter maturity** and **larger coupons**, since both pull the duration down (larger coupons put more PV weight on earlier times).

Macaulay Duration Formula

For a coupon bond:

$$D = \frac{1+\frac{\lambda}{m}}{\lambda} - \frac{1+\frac{\lambda}{m}+n\left(\frac{c}{m}-\frac{\lambda}{m}\right)}{c\left[\left(1+\frac{\lambda}{m}\right)^n - 1\right]+\lambda}$$

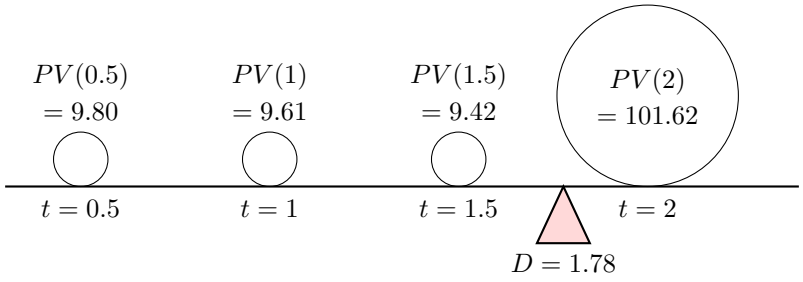
c : coupon rate per year
λ : yield
m : periods per year
n : periods remaining

Ex. A 2-year bond with face value \$100, a 20% semi-annual coupon (\$10 every 6 months), and a yield of 4% compounded semi-annually (2% per period). Find the Macaulay duration.
Total present value:

$$PV_{total} = \sum_{i=1}^{4} \frac{10}{(1+0.02)^i} + \frac{100}{(1+0.02)^4} = 9.804 + 9.612 + 9.423 + 9.238 + 92.385 = 130.462$$

Macaulay duration:

$$D = \underbrace{0.5}_{t_1} \cdot \frac{9.804}{130.462} + \underbrace{1.0}_{t_2} \cdot \frac{9.612}{130.462} + \underbrace{1.5}_{t_3} \cdot \frac{9.423}{130.462} + \underbrace{2.0}_{t_4} \cdot \frac{9.238+92.385}{130.462} = 1.777 \text{ years}$$



Think of the PVs as weights on a seesaw over the timeline: the "balance point" is at $t = 1.78$, pulled close to $t = 2$ by the large final payment ($PV(2) = 101.62$). The formula above gives the same answer with $c = 0.2$, $\lambda = 0.04$, $m = 2$, $n = 4$.
(Equivalently, the yield is $2\ln(1.02) = 3.9605\%$ continuously compounded.)

Modified Duration

Why not calculate the sensitivity of a bond to the yield exactly?

Def. **Modified Duration**

Let P be the price of a bond. The modified duration D_m of the bond is

$$D_m = -\frac{1}{P(\lambda_0)} \left.\frac{dP(\lambda)}{d\lambda}\right|_{\lambda=\lambda_0}$$

λ_0 : the current yield

$P(\lambda_0)$: the current price of the bond

- Roughly, think **sensitivity = derivative**.
- The minus sign is so it won't be a negative quantity (price falls as yield rises).
- Dividing by P makes it a **relative** sensitivity.

Modified duration gives the percentage change in price for a change in yield:

$$D_m \approx -\frac{1}{P}\frac{\Delta P}{\Delta \lambda} \quad \Rightarrow \quad \boxed{\Delta P \approx -D_m P \Delta\lambda}$$

Ex. If the yield increases by 1%, a bond with modified duration 5 drops in price by about 5%. With a current price of \$100.

$$\Delta P \approx -D_m P \Delta\lambda = -5(\$100)(0.01) = -\$5$$

Relationship between Modified and Macaulay Duration

$$D_m = \frac{D}{1+\frac{\lambda}{m}}$$

D_m : modified duration

D : Macaulay duration

λ : yield

m : number of compounding periods per year

(i) They differ only by a **constant factor**.

(ii) With continuous compounding ($m \to \infty$), $D_m \to D$.

Proof. **Setup.** Suppose there are m periods per year, the yield is λ, and the bond pays cash flow c_k at period k (time k/m years), for $k = 0, 1, \ldots, n$. Then

$$PV_k = \frac{c_k}{\left(1+\frac{\lambda}{m}\right)^k}, \qquad \underbrace{PV_{tot}}_{\text{Current price of bond}} = \sum_{k=0}^{n} PV_k = \sum_{k=0}^{n} \frac{c_k}{\left(1+\frac{\lambda}{m}\right)^k}$$

and the two quantities we want to relate are

$$D = \sum_{k=0}^{n} \frac{k}{m} \frac{PV_k}{PV_{tot}} \quad \text{(Macaulay)}, \qquad D_m = -\frac{1}{PV_{tot}} \frac{dPV_{tot}}{d\lambda} \quad \text{(modified)}$$

Step 1: differentiate the price. Only $\left(1+\frac{\lambda}{m}\right)^{-k}$ depends on λ; by the chain rule,

$$\frac{d}{d\lambda}\left(1+\frac{\lambda}{m}\right)^{-k} = -k\left(1+\frac{\lambda}{m}\right)^{-k-1} \cdot \frac{1}{m}$$

so differentiating PV_{tot} term by term gives

$$\frac{dPV_{tot}}{d\lambda} = -\sum_{k=0}^{n} \frac{k}{m} \frac{c_k}{\left(1+\frac{\lambda}{m}\right)^{k+1}}$$

Step 2: rewrite in terms of PV_k. Pull one factor of $\left(1+\frac{\lambda}{m}\right)$ out of each denominator:

$$\frac{dPV_{tot}}{d\lambda} = -\frac{1}{1+\frac{\lambda}{m}} \sum_{k=0}^{n} \frac{k}{m} \underbrace{\frac{c_k}{\left(1+\frac{\lambda}{m}\right)^{k}}}_{PV_k} = -\frac{1}{1+\frac{\lambda}{m}} \sum_{k=0}^{n} \frac{k}{m} PV_k$$

Step 3: plug into the definition of D_m.

$$\begin{aligned} D_m &= -\frac{1}{PV_{tot}} \frac{dPV_{tot}}{d\lambda} \\ &= -\frac{1}{PV_{tot}} \left(-\frac{1}{1+\frac{\lambda}{m}} \sum_{k=0}^{n} \frac{k}{m} PV_k \right) \\ &= \frac{1}{1+\frac{\lambda}{m}} \underbrace{\sum_{k=0}^{n} \frac{k}{m} \frac{PV_k}{PV_{tot}}}_{D} \\ &= \frac{D}{1+\frac{\lambda}{m}} \end{aligned}$$

□

EXAM NOTE! remember how to calculate the **DERIVATIVE!** $\left.\frac{dP(\lambda)}{d\lambda}\right|_{\lambda=\lambda_0}$

Ex. A 10%, 30-year bond priced at \$100 has Macaulay duration $D = 9.94$, with semi-annual coupons.

(a) Modified duration, and the effect of a 1% yield increase: since the price is \$100 (par), the yield is $\lambda = 10\%$, so

$$D_m = \frac{D}{1 + \frac{\lambda}{2}} = \frac{9.94}{1.05} = 9.47$$

A 1% increase in yield causes about a 9.47% drop in price.

(b) Estimated price if the yield moves to 11%:

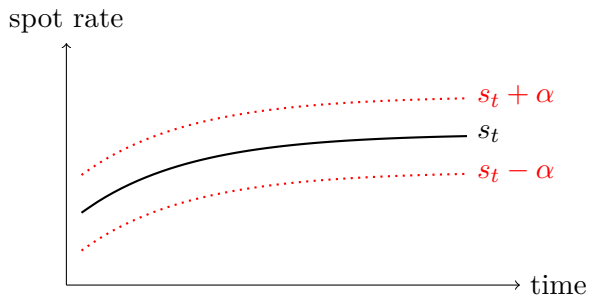
$$\Delta P \approx -D_m P \Delta\lambda = -9.47(100)(0.01) = -9.47 \quad \Rightarrow \quad P \approx \$100 - \$9.47 = \$90.53$$

Duration with Spot Rates

This time we compute sensitivity to changes in the **term structure** rather than the yield. It is again basically the derivative of the bond price with respect to interest rates, properly normalized, but now we perturb the entire spot rate curve.

Sensitivity to Spot Rates

Assume continuous compounding. What is the sensitivity of the present value of the cash flow $(x_{t_0}, x_{t_1}, \ldots, x_{t_n})$ to **parallel shifts** in the spot rate curve?
Note that its assumed all spot rates change at the same rate, hence parallel shift in the curve.



Shift the whole curve by α:

$$PV = \sum_{i=0}^{n} x_{t_i} e^{-s_{t_i} t_i} \quad \rightarrow \quad PV(\alpha) = \sum_{i=0}^{n} x_{t_i} e^{-(s_{t_i} + \alpha) t_i}$$

To determine sensitivity, **take the derivative with respect to α and evaluate at $\alpha = 0$**. Note that the current present value is $PV(0)$.

Def. **Duration with respect to Spot Rates**

$$D(s_t) = -\left(\frac{1}{PV(0)}\right)\frac{dPV(\alpha)}{d\alpha}\Big|_{\alpha=0}$$

current price

shift curve by α

price falls as rates rise $\left(\frac{dPV}{d\alpha}<0\right)$, so the minus makes $D(s_t)>0$

This is the **same recipe as modified duration**; only the thing we perturb changes:

$$\underbrace{D_m = -\frac{1}{P(\lambda_0)}\frac{dP(\lambda)}{d\lambda}\Big|_{\lambda=\lambda_0}}_{\text{perturb the yield } \lambda} \quad \longleftrightarrow \quad \underbrace{D(s_t) = -\frac{1}{PV(0)}\frac{dPV(\alpha)}{d\alpha}\Big|_{\alpha=0}}_{\text{perturb every spot rate by } \alpha}$$

- With **continuous compounding**, it is called the **Fisher-Weil duration** (it looks similar to Macaulay!):

$$D_{FW} = \left(\frac{1}{PV}\right)\sum_{i=0}^{n} t_i \underbrace{x_{t_i}e^{-s_{t_i}t_i}}_{PV_{t_i}}$$

time

since $\frac{dPV(\alpha)}{d\alpha}\Big|_{\alpha=0} = -\sum_{i=0}^{n} t_i x_{t_i} e^{-s_{t_i}t_i}$.

It is the same as the Macaulay duration, but generalized for all cash flow vectors, while Macaulay is for bonds.

If the spot curve is flat ($s_{t_i} = \lambda$ for all i), then $D_{FW} = D$.

- With **discrete compounding** (m periods per year, s_i the spot rate for period i), it is called the **quasi-modified duration**:

Note it is not the Macaulay duration, as there is a factor of $+1$ in the power of the denominator.

If the spot curve is flat ($s_i = \lambda$ for all i), then $D_Q = \frac{D}{1+\frac{\lambda}{m}} = D_m$.

Simplification for Continuous Compounding Only!

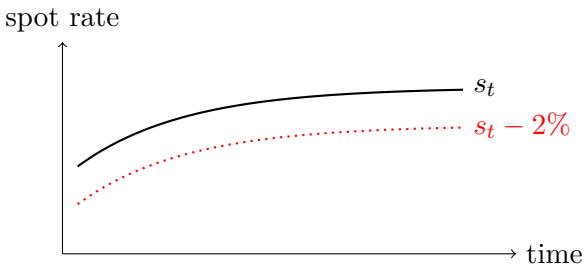
Under continuous compounding, all the duration formulas look like a **weighted average of times**:

$$D_{FW} = \sum_{i=0}^{n} t_i \frac{PV_{t_i}}{PV_{Total}}$$

The only difference is the rate used to compute the PVs:

- Macaulay and modified duration use the **yield**: $PV_{t_i} = x_{t_i}e^{-\lambda t_i}$
- Fisher-Weil duration uses the **spot rates**: $PV_{t_i} = x_{t_i}e^{-s_{t_i}t_i}$

Ex.



The entire spot rate curve drops by 2%. A bond with Fisher-Weil duration 10 changes in price by approximately $-D_{FW}\Delta\alpha = -10(-0.02) = +20\%$, i.e. it **increases by 20%**.

Summary

Macaulay duration is a weighted average of the times to cash flows:

$$D = w_{t_1}t_1 + w_{t_2}t_2 + \cdots + w_{t_n}t_n, \qquad w_{t_i} = \frac{PV(t_i)}{PV_{total}}$$

The other durations measure a percentage change in price for a change in either the yield or a parallel shift of the spot rate curve.

- **Modified duration** uses the yield, and is related to Macaulay duration by

$$D_m = -\frac{1}{P(\lambda_0)}\left.\frac{dP(\lambda)}{d\lambda}\right|_{\lambda=\lambda_0}, \qquad D_m = \frac{D}{1+\frac{\lambda}{m}}$$

- **Fisher-Weil** (continuous compounding) and **quasi-modified** (discrete compounding) use the spot rate curve:

$$D(s_t) = -\left(\frac{1}{PV(0)}\right)\left.\frac{dPV(\alpha)}{d\alpha}\right|_{\alpha=0}$$

$$D_{FW} = \left(\frac{1}{PV}\right)\sum_{i=0}^{n} t_i x_{t_i} e^{-s_{t_i}t_i}, \qquad D_Q = \left(\frac{1}{PV}\right)\sum_{i=1}^{n}\frac{\frac{i}{m}x_i}{(1+\frac{s_i}{m})^{i+1}}$$

Lecture 7: Duration for Bond Portfolios & Immunization

Oct 8, 2026

Duration of a Portfolio

Def. **Portfolio Duration**

Given fixed income securities with prices P_i and durations D_i, $i = 1, \ldots, m$, the portfolio consisting of the aggregate of these has price P and duration D:

$$P = P_1 + P_2 + \cdots + P_m = \sum_{i=1}^{m} P_i$$

$$D = w_1 D_1 + w_2 D_2 + \cdots + w_m D_m, \quad w_i = \frac{P_i}{P} \quad 0 < w < 1, \sum w_i = 1$$

Works for both **Macaulay** and **all forms of modified** duration.

Note: when having multiple (x_i) shares of one bond

$$P = x_1 P_1 + x_2 P_2 + \cdots + x_m P_m = \sum_{i=1}^{m} x_i P_i$$

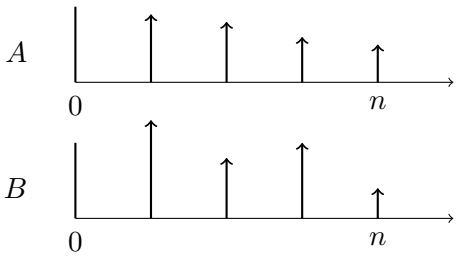
$$D = w_1 D_1 + w_2 D_2 + \cdots + w_m D_m =, \qquad w_i = \frac{x_i P_i}{P}$$

Note: this looks just like the Macaulay duration formula, except with **time replaced by duration**!

(Weights are each security's share of the total price.)

Derivation

Take two cash flow streams A and B over the same times $t_0, \ldots, t_n$ (a stream with no payment at t_k just has $PV_k = 0$ there):



$$D^A = \sum_{k=0}^{n} \frac{t_k PV_k^A}{P^A}$$

$$D^B = \sum_{k=0}^{n} \frac{t_k PV_k^B}{P^B}$$

The duration of $A + B$ (cash flows add, so PVs add):

$$D^{A+B} = \sum_{k=0}^{n} \frac{t_k \left(PV_k^A + PV_k^B\right)}{P^A + P^B} = \sum_{k=0}^{n} \frac{t_k PV_k^A}{P^A + P^B} + \sum_{k=0}^{n} \frac{t_k PV_k^B}{P^A + P^B}$$

$$= \sum_{k=0}^{n} \frac{t_k PV_k^A}{P^A + P^B} \cdot \boxed{\frac{P^A}{P^A}} + \sum_{k=0}^{n} \frac{t_k PV_k^B}{P^A + P^B} \cdot \boxed{\frac{P^B}{P^B}} \qquad \text{(multiply by 1)}$$

$$= \underbrace{\sum_{k=0}^{n} \frac{t_k PV_k^A}{\cancel{P^A}}}_{D^A} \left(\frac{\cancel{P^A}}{P^A + P^B}\right) + \underbrace{\sum_{k=0}^{n} \frac{t_k PV_k^B}{\cancel{P^B}}}_{D^B} \left(\frac{\cancel{P^B}}{P^A + P^B}\right) \qquad \text{(swap denominators)}$$

$$\boxed{D^{A+B} = \left(\frac{P^A}{P^A + P^B}\right) D^A + \left(\frac{P^B}{P^A + P^B}\right) D^B = w_A D^A + w_B D^B}$$

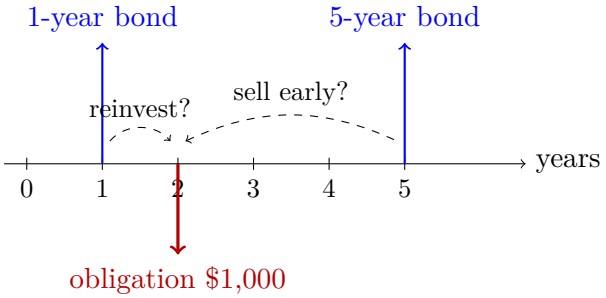
The weights add to 1 ($w_A + w_B = 1$), so D^{A+B} always lies between D^A and D^B (for positive prices). Repeating this for more streams gives the general formula.
Note: the derivation needs every PV_k to be computed with the same rate (one common yield, or one spot rate curve). If the bonds have different yields, the formula is only an approximation.

Immunization

Def. **Immunization**
Constructing a portfolio of bonds which is **protected (immunized) against changes in interest rates**.

Problem: you have an obligation to pay \$1,000 two years from now, but you can only purchase bonds of maturity 1 and 5 years.
Reasons to buy 5 year bond instead of 2 year: longer maturity, yield and prices may be better

- 1 year bond only: **reinvestment risk**. The money must be reinvested at year 1 at an unknown future rate.
- 5 year bond only: **too sensitive to interest rates**. It must be sold at year 2 at an unknown price.



Solution: construct a portfolio of 1 and 5 year securities with the same **present value** and **duration** as your obligation.
Ensures the prices of the bonds could grow to the magnitude of the liability.

With x units of bond 1 (price P_1, duration D_1) and y units of bond 2 (price P_2, duration D_2), and an obligation with present value P and duration D:

$$\begin{aligned} \text{PV:} \quad & xP_1 + yP_2 = P \\ \text{Duration:} \quad & \frac{xP_1}{P}D_1 + \frac{yP_2}{P}D_2 = D \end{aligned}$$

Solve these two equations for x and y to determine your portfolio. The duration equation is just the portfolio duration formula, with weights $w_1 = \frac{xP_1}{P}$ and $w_2 = \frac{yP_2}{P}$.

Why it works: equal PV means equal value now. Equal duration means equal $\frac{dP}{d\lambda}$ (since $\frac{dP}{d\lambda} = -D_m P$), so when the yield moves a little, the portfolio and the obligation change in value by about the same amount. The portfolio can still cover the obligation.

Ex. You have an obligation of \$1,000 in 2 years, and wish to invest now to meet it. You can invest in 1 year and 5 year zero-coupon bonds (face \$100, discrete yearly compounding, yield 9%):

	Coupon	Maturity	Yield	Price	Duration (Mac)
Obligation	\$1,000 in 2 years			\$841.68	2 years
Bond 1	0%	1 year	9%	\$91.74	1 year
Bond 2	0%	5 years	9%	\$64.99	5 years

Prices: $\frac{1000}{1.09^2} = 841.68$, $\frac{100}{1.09} = 91.74$, $\frac{100}{1.09^5} = 64.99$. Each is a zero, so its Macaulay duration is its maturity.
Note the price of the liability is discounted by the rate of the yield, when yields are different you can take an average (only an approximation).
Let x = number of bond 1, y = number of bond 2 held.

$$\begin{aligned} \text{Match prices:} \quad & 91.74x + 64.99y = 841.68 \\ \text{Match durations:} \quad & \frac{91.74x}{841.68}(1 \text{ year}) + \frac{64.99y}{841.68}(5 \text{ years}) = 2 \text{ years} \end{aligned}$$

Multiplying the duration equation by 841.68 and subtracting the price equation: $4(64.99)y = 841.68$, so

$$x = 6.88, \quad y = 3.24$$

In weights, $w_1 + w_2 = 1$ and $w_1(1) + w_2(5) = 2$ give $w_1 = \frac{3}{4}$ and $w_2 = \frac{1}{4}$. So \$631.26 goes into bond 1 and \$210.42 into bond 2.
Sell all of them at the end of duration!

Ex. Bond Immunization Spreadsheet
Yield 9% (semi-annual compounding, 4.5% per period). Obligation of \$1,000,000 in 10 years, to be immunized using:

	Maturity	Principal	Coupon	Price	Macaulay D	\$ amount
Bond 1	20	\$1,000	4%	\$539.96	11.45191	\$321,333.61
Bond 2	5	\$1,000	0%	\$643.93	5	\$93,309.25
Obligation	10	\$1,000,000	0%	\$414,642.86	10	\$414,642.86

- Prices at 4.5% per period, e.g. bond 2: $\frac{1000}{1.045^{10}} = 643.93$, obligation: $\frac{1{,}000{,}000}{1.045^{20}} = 414{,}642.86$.
- Match durations with weights: $11.45191\, w_1 + 5(1 - w_1) = 10 \;\Rightarrow\; w_1 = \frac{5}{6.45191} = 0.775$.
- \$ amounts: $0.775(414{,}642.86) = \$321{,}333.61$ in bond 1, and the rest, \$93,309.25, in bond 2.
- Number of bonds $= \frac{\text{\$ amount}}{\text{price}}$: **595.1059** of bond 1 and **144.9064** of bond 2.

Note the target duration (10) must lie between the two bonds' durations (5 and 11.45). Otherwise one weight is negative, i.e. a short position.

The Real Idea Behind Immunization: Taylor Expansions!

With λ_0 the current value of the yield, expand the price as a function of the yield around the current value:

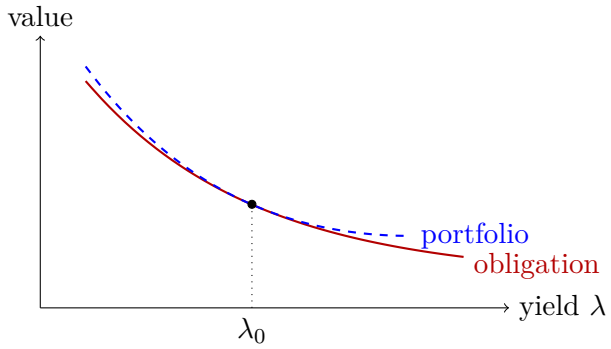
$$P(\lambda) = \underbrace{P(\lambda_0)}_{\text{price}} + \underbrace{P'(\lambda_0)(\lambda - \lambda_0)}_{\text{duration}} + \underbrace{\tfrac{1}{2}P''(\lambda_0)(\lambda - \lambda_0)^2}_{\text{convexity}} + \cdots$$

Here duration and convexity are **unnormalized**: $P'(\lambda_0) = -D_m P(\lambda_0)$ is modified duration times price.

You can match as many terms as you like to make the Taylor series of your portfolio look like the Taylor series of your cash flow stream (obligation).

- Matching **PV and duration** (as above) matches the first two terms. The portfolio and obligation price-yield curves are **tangent** at λ_0.

- Matching **convexity** too would match the curvature, but needs a third equation, so a third bond.



Same value and slope at λ_0, so small yield changes affect both equally. For larger changes, the curvature (convexity) difference shows up.